Q19: What Are the Signs That AI Has Its Own Consciousness?

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Learning objectives

- Distinguish consciousness from intelligence, capability, and self-report.
- Summarize how leading scientific theories of consciousness have been applied to AI.
- Reason carefully about why behavioral "signs" are weak evidence, and what better evidence looks like.

Companion videos (watch in order):

- *Why AI Isn't Going to Become Conscious* — Anil Seth / TED (14:58). https://www.youtube.com/watch?v=tJV-vdbZ388
- *How could we tell whether AI has become conscious?* — Sabine Hossenfelder (10:23). https://www.youtube.com/watch?v=CSTfgYynziw

1 First, separate the concepts

Student response #01 asks what the signs are that an AI has its own consciousness. The first discipline is to *not* conflate consciousness with things it is routinely confused for: **intelligence** (solving hard problems), **capability** (what a system can do), **agency** (pursuing goals), and **self-report** (saying "I am conscious"). A system can have any of these without phenomenal experience, and—at least in principle—experience could exist without high intelligence. "Consciousness" here means *phenomenal* consciousness: that there is something it is like to be the system [3]. This is distinct from *access* consciousness (information being available for reasoning and report), which is more tractable but not the same question.

2 Why self-report is nearly worthless as evidence

An LLM trained on human text will fluently produce first-person claims of feelings because that is what the training distribution contains—not because it has introspective access to inner states. The report is *caused by* the data and the prompt, not by an experience. Worse, these claims are steerable: the same model will assert and deny consciousness depending on the prompt. So "it said it was conscious" is close to zero evidence. This is the single most important point for a student to internalize.

3 Theory-driven indicators

A more principled approach asks what each scientific theory of consciousness *requires*, then checks whether an architecture has it [1]. Examples:

- **Global Workspace Theory**: a bottleneck that broadcasts selected information globally. One can ask whether a model has a functional global workspace (some do, partially, by design).
- **Higher-Order theories**: representations *about* the system's own representations.
- **Integrated Information Theory** (IIT): consciousness tracks a system's irreducible integrated information Φ [4]. Computing Φ for real networks is intractable, and IIT makes the sharp, counterintuitive claim that a feed-forward network has $\Phi = 0$ regardless of behavior.
- **Recurrent / predictive-processing** accounts: emphasize recurrence and self-modeling.

Butlin et al. [2023] turn these into a checklist of "indicator properties" and conclude current systems satisfy few of them, but that there is no obvious technical barrier to building systems that satisfy more.

4 Why this is genuinely hard

Three deep problems block a clean test. (1) The **other-minds problem**: we infer consciousness in other humans by analogy from shared biology and behavior—an AI breaks that analogy. (2) **Gaming**: a capable system can produce any behavioral "sign" we name, so behavioral tests are not robust to a system optimized to pass them. (3) **Theory disagreement**: the theories above disagree with each other, so an "indicator" is only as trustworthy as a theory we have not confirmed. The responsible scientific posture is therefore probabilistic and theory-relative, not a yes/no verdict.

5 A concrete walk through one theory

To make "theory-driven indicator" concrete, take Global Workspace Theory (GWT). GWT holds that a mental state is conscious when its content is *globally broadcast*: selected by an attentional bottleneck and made available to many downstream processes (memory, reasoning, report) [2]. Applied to AI, the indicator question becomes architectural and checkable: does the system have a limited-capacity workspace that gates information, and is the gated content broadcast to otherwise independent modules? A vanilla transformer has attention (a selection mechanism) but not a persistent capacity-limited workspace broadcasting to specialized subsystems, so it scores *partially* on GWT indicators—and one could deliberately build a system that scores higher. Crucially, scoring high on GWT indicators would make the system "conscious according to GWT," which is only meaningful if GWT is correct—a question neuroscience has not settled. This is why serious work reports indicator scores *per theory* with explicit uncertainty, rather than a single verdict, and why Rick's machine-consciousness program runs sham-controlled contrasts tied to specific theories (global workspace, recurrence, higher-order, integrated-information proxies) instead of asking the model how it feels.

6 The moral stakes of getting it wrong

Two asymmetric errors loom. A *false positive*—attributing experience to a system that has none—could paralyze useful work and misdirect moral concern. A *false negative*—denying experience to a

system that has it—risks real moral harm at scale. Because the theories disagree and the evidence is thin, the responsible stance treats the question as a live uncertainty with non-trivial probability mass on several answers, and separates it cleanly from capability and welfare questions that can be addressed independently. Students should leave able to hold "we do not know" without collapsing it into either "obviously conscious" or "obviously just a program."

7 What better evidence would look like

Convergent evidence across independent theories (a system scoring high on global-workspace, higher-order, *and* recurrence indicators), causal-mechanistic tests (interventions that manipulate the putative substrate and change the putative markers), and architectures where the markers were *not* optimized for—so passing them is informative. Even then, the conclusion is "consistent with" not "proves." This is an active research area; Rick's own machine-consciousness program runs exactly these theory-grounded, sham-controlled contrasts rather than trusting self-report.

8 Summary

Consciousness is not intelligence, capability, agency, or self-report; self-report in LLMs is almost worthless because it is a trained behavior. The scientific route is theory-driven indicator properties (global workspace, higher-order, IIT, recurrence), but theories disagree, behavioral signs can be gamed, and the other-minds problem is real. The honest answer: there is no single reliable "sign," only theory-relative, probabilistic, convergent evidence—and current systems satisfy few indicators.

References

[1] Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, et al. Consciousness in artificial intelligence: Insights from the science of consciousness. *arXiv:2308.08708*, 2023.

[2] Stanislas Dehaene. *Consciousness and the Brain.* Viking, 2014.

[3] Thomas Nagel. What is it like to be a bat? *The Philosophical Review*, 83(4), 1974.

[4] Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: From consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17, 2016.